

Addition and subtraction of algebraic fractions



1

a $\frac{9x}{7}$

b $\frac{5x}{4}$

c $4x$

d $\frac{5x}{4}$

e $\frac{-4x}{9}$

f $\frac{-5x}{14}$

g $\frac{5x}{2}$

h $\frac{5x}{3}$

i $-\frac{3x}{5}$

j $4x$

k $\frac{5x}{6}$

l $-\frac{9x}{7}$

2

a $x + 1$

b $\frac{6x + 1}{4}$

c $\frac{2x - 3}{7}$

d $\frac{3x - 4}{3}$

e $-\frac{x}{3}$

f $\frac{-9x - 13}{5}$

g $\frac{x^2 + 4x + 11}{6}$

h $\frac{5x^2 + 3}{5}$

i $\frac{-8x^2 - 4x}{3}$

j $\frac{-4x^3 + x^2 + 2x - 1}{4}$

3

a 6

b 6

c 6

d 15

4

a 30

b $\frac{9m}{10}$

5 Example answers:

$$\frac{3x}{5} + \frac{2x}{7} = \frac{31x}{35}, \quad \frac{3x}{5} + \frac{2x}{35} = \frac{23x}{35}, \text{ or } \frac{3x}{5} + \frac{2x}{70} = \frac{22x}{35}$$

6

a $\frac{34x}{35}$

b $\frac{17x}{21}$

c $\frac{23x}{35}$

d $\frac{x}{30}$

e $\frac{-68x}{7}$

f $\frac{11x}{35}$

g $\frac{-27x}{8}$

h $\frac{47x}{42}$

i $\frac{16x}{105}$

j $-\frac{7x^2}{6}$

7

$$\text{a} \quad \frac{8x+5}{9}$$

$$\text{e} \quad \frac{30x+3}{44}$$

$$\text{i} \quad \frac{-x+19}{12}$$

$$\text{m} \quad \frac{3x+1}{8}$$

$$\text{b} \quad \frac{27x+4}{39}$$

$$\text{f} \quad -\frac{5y+2}{50}$$

$$\text{j} \quad \frac{4x+7}{12}$$

$$\text{n} \quad \frac{15x+26}{9}$$



$$\text{c} \quad \frac{4x-15}{80}$$

$$\text{g} \quad \frac{25x-12}{40}$$

$$\text{k} \quad \frac{4x+7}{3}$$

$$\text{o} \quad \frac{19x^2+5x}{21}$$

$$\text{d} \quad \frac{6x-1}{12}$$

$$\text{h} \quad \frac{7x+10}{12}$$

$$\text{l} \quad \frac{33x+44}{30}$$

$$\text{p} \quad \frac{11x^2+2x}{28}$$

8

$$\text{a} \quad \frac{5x+3y}{xy}$$

$$\text{b} \quad \frac{2a+c^3}{abc^2}$$

$$\text{c} \quad \frac{2x^3+3xy+3y}{xy}$$

$$\text{d} \quad \frac{6x-8}{x^2-2x}$$